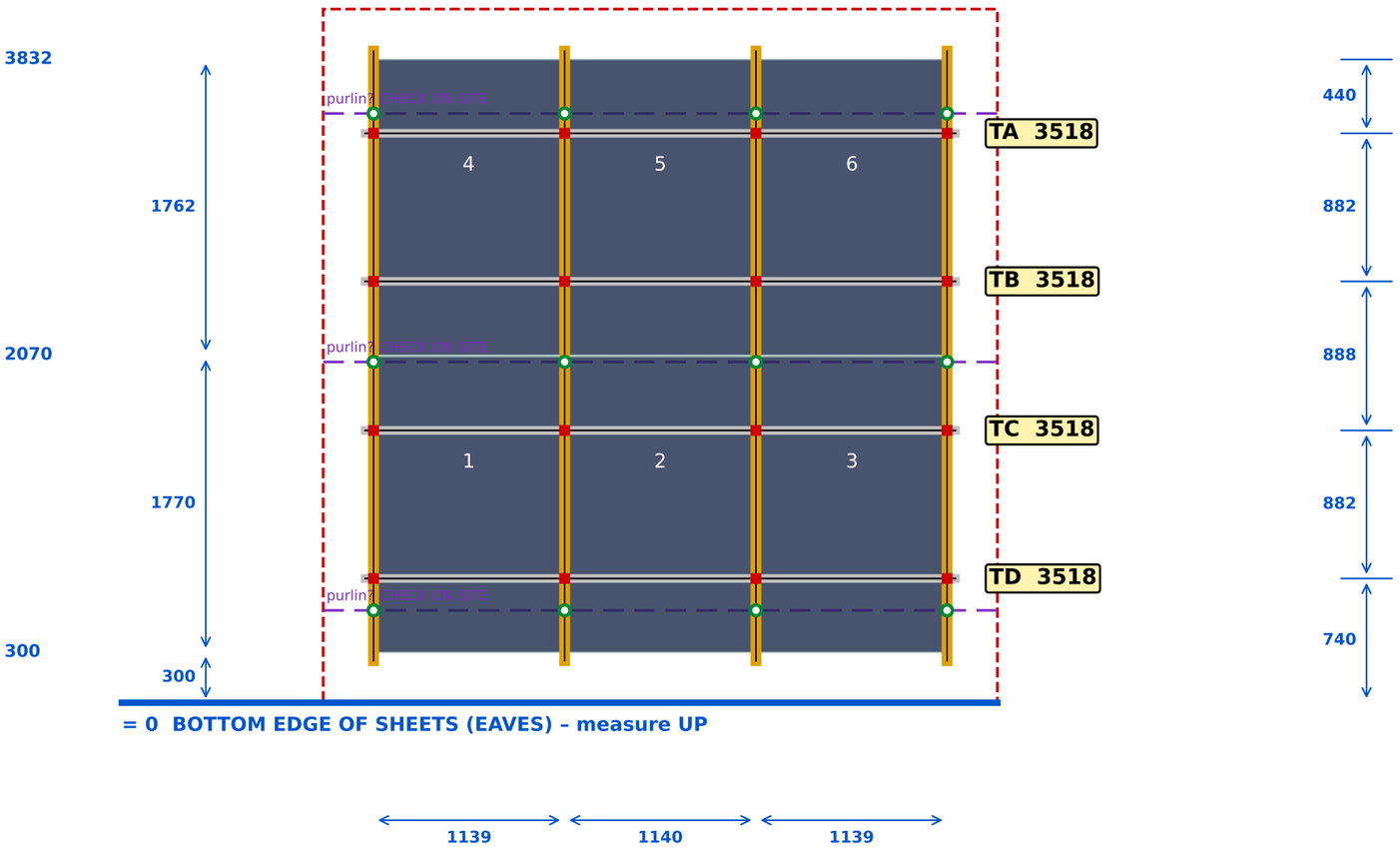
FINAL – 12 PANELS OUTBUILDING – PORTRAIT ON OVERLAY RAILS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | OUTBUILDING (12) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v6

2 faces × 6 | PORTRAIT 2 rows × 3 | array 3418 wide × 3532 up slope | BASE rails UP the roof on purlins, TOP rails ALONG roof on K2 cross connectors

Top of panels 3832 • RIDGE ≥ 4132 – CHECK ON SITE



= 0 BOTTOM EDGE OF SHEETS (EAVES) – measure UP

BASE rails B1…B4 across (centres from left panel edge) – blue mm

Orange = BASE rails UP the roof (3632, B1…B4), fixed at EVERY purlin (green ○ = 2003016 + 2002544). Grey = TOP rails ALONG roof (TA top…TD bottom), 2 per row, 440 in from panel ends. Red ■ = K2 cross connector (part – ring Dale to confirm). Red tick = joiner. SAME BOTH FACES. Purple = purlins (nominal).

SET-OUT (up from = 0 at bottom edge of sheets)

Panels start 300. Row 1 300–2062, 8 gap, row 2 2070–3832. Base rails 250 → 3882.

Top rails TD 740 TC 1622 TB 2510 TA 3392 (gaps 740, 882, 888, 882, 440)

Slope needed ≥ 4132 vs ~3,800 estimate → 332 SHORT. Width ≥ 4018. CHECK ON SITE

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

NO BIRD MESH ON THIS JOB

FIXING – K2 2003016 Solarfastener 8×155/50 (STEEL)

1. Mark on CREST over purlin centre. Measure flange thickness.
2. **Sheet hole 11 mm, NO hammer. Steel 1.5–3 mm: self-drills. Thicker steel: pilot 6.8–7.4 mm. Clamp range 84–125 mm.**
3. Screw in at 90°, seal just compressed – don't over-drive.
4. 2002544 M10 adapter plate (1 per fastener) + T-bolt 1000571 + flange nut 1000042. Torque 32 Nm rail / 35 Nm joiners.
5. Fix every 1.0–1.2 m along each purlin.

PER FACE

BASE: 4 × 3632 up slope (1 bar each, ~1139 apart)

TOP: 4 × 3518 = 3518

Fixings 12 (4 × 3 purlins) Cross conn. 16

End clamps 8 Mid clamps 8 Joiners 0

LOAD LIST – 12 PANELS (both faces)

- ☐ **12 × JA JAM54D40-460/LB**
- ☐ **16 × K2 rail 3.65 m (8 base + 0 full top + 8 for 8×3518)**
- ☐ **0 × joiners ☐ 16 × end 30 mm ☐ 16 × mid 8 mm**
- ☐ **32 × K2 cross connectors – not delivered – ring Dale to confirm**
- ☐ **24 × 2003016 fasteners (have 120 → 96 spare) ✓**
- ☐ **24 × 2002544 plates (have 85 → 61 spare) ✓**
- ☐ **24 × T-bolt 1000571 + 24 × flange nut 1000042**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits
☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm
☐ Hacksaw + 24/32 TPI blades (alu), file/deburr
☐ Tape, chalk line, marker, masking tape, square
☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties
☐ Sealant gun + sealant
☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes
☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes
☐ Crawl boards, harness/lanyards, edge protection
☐ **NO electrical tools – electrics are Regen's**

FLAGS – CHECK ON SITE

1. Count purlins each base rail crosses (3 assumed; 4 → 32 fixings).
2. Slope: needs ≥ 4132 vs est. 3,800 → ~330 SHORT: may not fit 300 mm rule.
3. Cross connectors not on delivery list – ring Dale to confirm before going up.
4. Flange > 3 mm → pilot 6.8–7.4 mm.

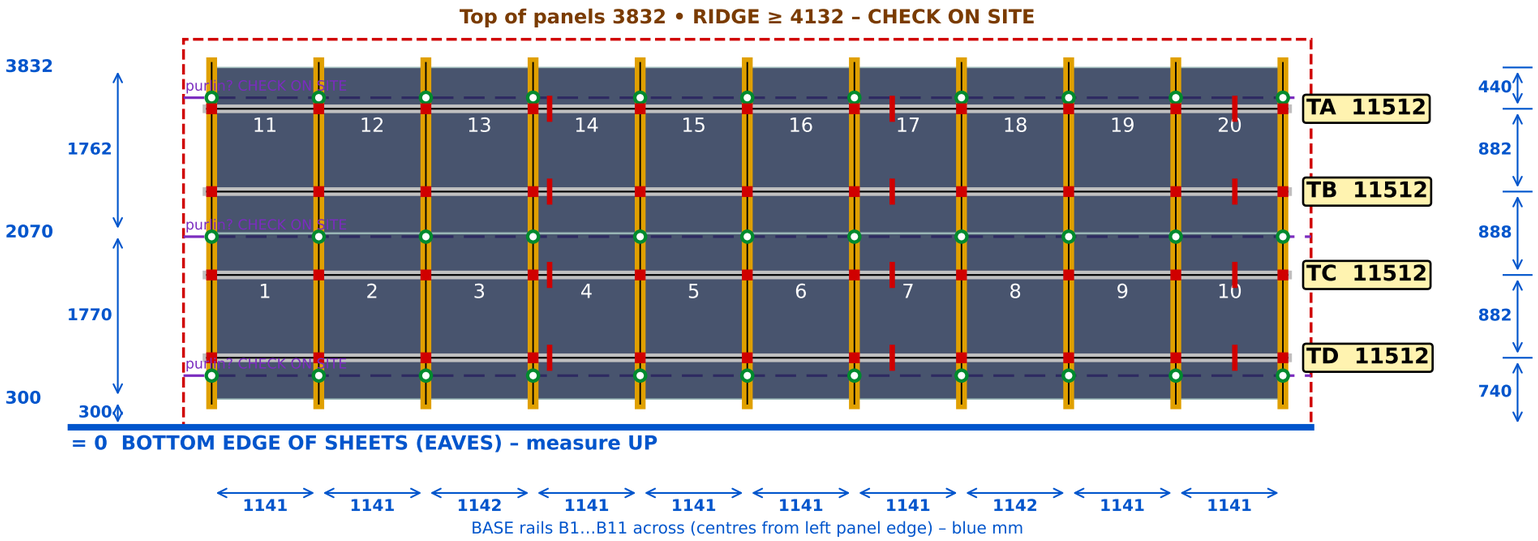
Top-rail cut bars: 3518 | 3518 | 3518 | 3518 | 3518 | 3518 | 3518 | 3518

Not for upload – John's info only. Not a structural design. Purlin-dependent dims marked CHECK ON SITE.

FINAL – 40 PANELS MAIN BARN – PORTRAIT ON OVERLAY RAILS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | MAIN BARN (40) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v6

2 faces × 20 | PORTRAIT 2 rows × 10 | array 11412 wide × 3532 up slope | BASE rails UP the roof on purlins, TOP rails ALONG roof on K2 cross connectors



Orange = BASE rails UP the roof (3632, B1…B11), fixed at EVERY purlin (green ○ = 2003016 + 2002544). Grey = TOP rails ALONG roof (TA top…TD bottom), 2 per row, 440 in from panel ends. Red ■ = K2 cross connector (part – ring Dale to confirm). Red tick = joiner. SAME BOTH FACES. Purple = purlins (nominal).

SET-OUT (up from = 0 at bottom edge of sheets)

Panels start 300. Row 1 300–2062, 8 gap, row 2 2070–3832. Base rails 250 → 3882.

Top rails TD 740 TC 1622 TB 2510 TA 3392 (gaps 740, 882, 888, 882, 440)

Slope needed ≥ 4132 vs ~3,800 estimate → 332 SHORT. Width ≥ 12012. CHECK ON SITE

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

NO BIRD MESH ON THIS JOB

FIXING – K2 2003016 Solarfastener 8×155/50 (STEEL)

1. Mark on CREST over purlin centre. Measure flange thickness.
2. **Sheet hole 11 mm, NO hammer. Steel 1.5–3 mm: self-drills. Thicker steel: pilot 6.8–7.4 mm. Clamp range 84–125 mm.**
3. Screw in at 90°, seal just compressed – don't over-drive.
4. 2002544 M10 adapter plate (1 per fastener) + T-bolt 1000571 + flange nut 1000042. Torque 32 Nm rail / 35 Nm joiners.
5. Fix every 1.0–1.2 m along each purlin.

PER FACE

BASE: 11 × 3632 up slope (1 bar each, ~1141 apart)

TOP: 4 × 11512 = 3650 + 3650 + 3650 + 562

Fixings 33 (11 × 3 purlins) Cross conn. 44

End clamps 8 Mid clamps 36 Joiners 12

LOAD LIST – 40 PANELS (both faces)

- ☐ **40 × JA JAM54D40-460/LB**
- ☐ **48 × K2 rail 3.65 m (22 base + 24 full top + 2 for 8×562)**
- ☐ **24 × joiners ☐ 16 × end 30 mm ☐ 72 × mid 8 mm**
- ☐ **88 × K2 cross connectors – not delivered – ring Dale to confirm**
- ☐ **66 × 2003016 fasteners (have 120 → 54 spare) ✓**
- ☐ **66 × 2002544 plates (have 85 → 19 spare) ✓**
- ☐ **66 × T-bolt 1000571 + 66 × flange nut 1000042**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits
☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm
☐ Hacksaw + 24/32 TPI blades (alu), file/deburr
☐ Tape, chalk line, marker, masking tape, square
☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties
☐ Sealant gun + sealant
☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes
☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes
☐ Crawl boards, harness/lanyards, edge protection
☐ **NO electrical tools – electrics are Regen's**

FLAGS – CHECK ON SITE

1. Count purlins each base rail crosses (3 assumed; 4 → 88 fixings).
2. Slope: needs ≥ 4132 vs est. 3,800 → ~330 SHORT: may not fit 300 mm rule.
3. Cross connectors not on delivery list – ring Dale to confirm before going up.
4. Flange > 3 mm → pilot 6.8–7.4 mm.

Top-rail cut bars: 562+562+562+562+562+562 | 562+562

Not for upload – John's info only. Not a structural design. Purlin-dependent dims marked CHECK ON SITE.

FINAL LOAD LIST – PORTRAIT OVERLAY – 12 + 40 PANELS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | MAIN BARN (40) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v6

Both systems: base rails UP roof on purlins (3 purlins assumed – CHECK ON SITE), top rails ALONG roof on K2 cross connectors

Item	12-panel	40-panel	TOTAL
JA JAM54D40-460/LB panels	12	40	**52**
K2 rail 3.65 m bars	16	48	**64**
Rail joiners	0	24	**24**
End clamps 30 mm	16	16	**32**
Mid clamps 8 mm	16	72	**88**
K2 cross connectors (ring Dale to confirm)	32	88	**120**
2003016 Solarfastener 8×155/50	24	66	**90**
2002544 M10 adapter plates	24	66	**90**
M10 T-bolt 1000571	24	66	**90**
Flange nut 1000042	24	66	**90**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits

☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm

☐ Hacksaw + 24/32 TPI blades (alu), file/deburr

☐ Tape, chalk line, marker, masking tape, square

☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties

☐ Sealant gun + sealant

☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes

☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes

☐ Crawl boards, harness/lanyards, edge protection

☐ **NO electrical tools – electrics are Regen's**

vs DELIVERED

Fasteners: need 90, have 120 → +30

Plates: need 90, have 85 → -5

(delivery is for the 40 only? If 12-panel kit separate, 40-panel alone: 66 vs 120 / 85)

Cross connectors 120 – not on delivery list – ring Dale to confirm

Plates short – ring Dale to confirm

NO BIRD MESH ON THIS JOB

Rails: 12-panel top 4 × 3518 (3518), base 8 × 3632; 40-panel top 8 × 11512 (3650+3650+3650+562), base 22 × 3632

40-panel slope needs ≥ 4132 vs ~3,800 est. – CHECK ON SITE

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

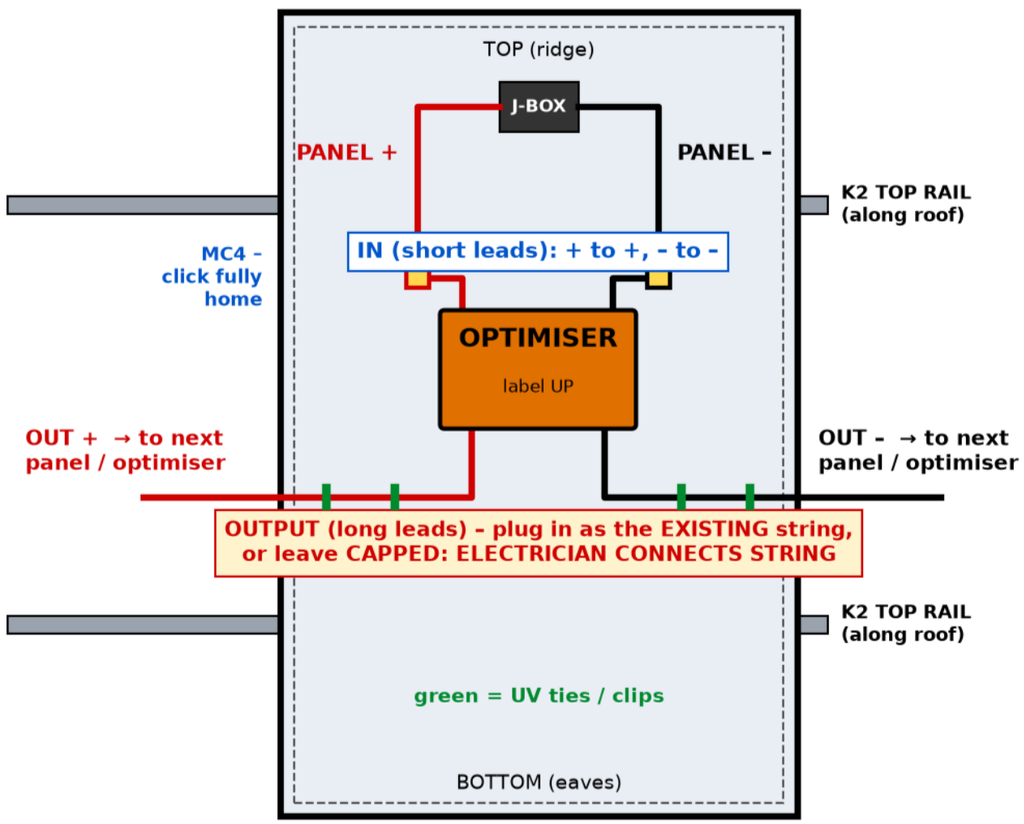
OPTIMISER FITTING – 2 optimisers on the 12-panel OUTBUILDING array

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | JA JAM54D40-460/LB PORTRAIT 1762×1134×30 on K2 overlay rails | v1

PHYSICAL FITTING + PLUG-IN ONLY. No string design, no inverter wiring – ELECTRICIAN CONNECTS THE STRING.

REAR VIEW – underside of one portrait panel

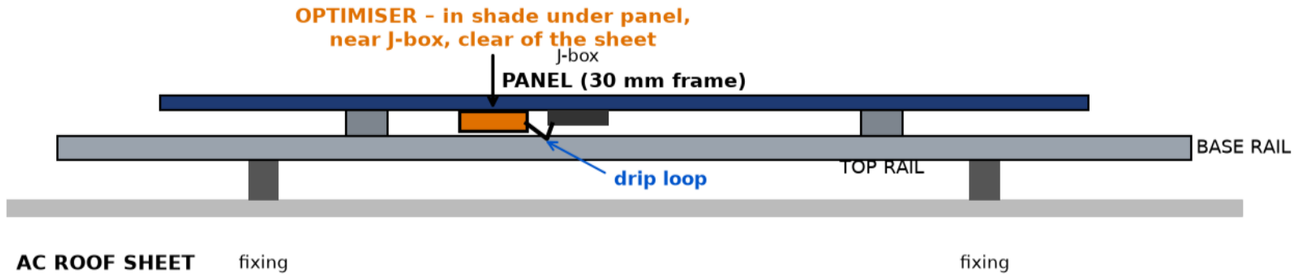
(looking at the back, top of panel = ridge end)



Cables tied to rail/frame – NEVER lying on the roof sheet.
No tension, no tight bends, drip loop at each plug.

SIDE VIEW – across the slope

Gap under optimiser to roof sheet – rain must run off freely.
Never sit it on the sheet or in a valley/trough.



CONFIRM ON THE BOX FIRST

- ☐ Optimiser make / model
- ☐ WHICH 2 panels – ask Regen / electrician (mark on layout)
- ☐ Mounting: frame clip OR rail bracket + bolt
- ☐ Connectors MC4 – same as panel (don't mix brands, NEVER cut)
- ☐ Which leads are IN vs OUT

⚠ SAFETY – PANELS ARE LIVE

- In daylight panels make DC – even covered, treat as live
- **NEVER pull a connector under load** (string must be open / isolated)
- Don't cut MC4s, don't strip leads
- Push till it CLICKS, tug test
- Anything odd → STOP, ring sparky

STEPS (plain English)

1. Check the box (red list). Mark the 2 panels on the layout sheet.
2. Peel the spare SERIAL sticker off each optimiser onto the layout sheet at that panel's spot (photo the serial too).
3. Fix optimiser to the panel frame / top rail BEFORE the panel goes down – under the panel, near the J-box, label up, off the roof sheet.
4. Plug panel + into optimiser IN +, panel – into IN –. Click home, tug test.
5. OUT leads: plug on as the existing string, or cap & leave – ELECTRICIAN CONNECTS THE STRING.
6. Tie all cables to rail/frame (UV ties), drip loops, no tension, nothing on the sheet. Then clamp the panel down.